Rules of probability

Outcomes, sample space, events

In probability theory, we make mathematical models to describe chance experiments. For a chance experiment, we identify a set of possible outcomes, which we call the sample space Ω for the experiment.

A good example to keep in mind is the chance experiment where we roll two dice, a red die and a green die. Durrett gives a sample space Ω to model this experiment in Example 1.2. He describes outcomes as ordered pairs like (1, 2) and (4, 6). The first number in the ordered pair is the number we roll on the red die, and the second number is the number we roll on the green die.

When we think about a chance experiment, we are often interested not just in specific outcomes but also in more complex **events**. An event is a collection of outcomes. In more technical mathematical terminology, we say that an event is a subset of the sample space Ω. In Durrett's dice-rolling situation, examples of events include "the red roll is bigger than the green roll" (with 15 outcomes) and "the red roll and the green roll differ by at most 1" (with 16 outcomes).

Basic rules of probability (minus one)

When we set up a mathematical model to reason about a chance experiment, we specify a sample space Ω and a way to assign a probability $P(A)$ to each event. The probability assignment must follow these rules:

- For any event A, the probability $P(A)$ that A happens is a real number between 0 and 1.
- $P(\Omega) = 1$
- If A and B are disjoint events, then $P(A \cup B) = P(A) + P(B)$.
- There is one more rule necessary for sample spaces with infinitely many outcomes. You have read about this rule in Durrett, and we will discuss it later in the semester.

Equally likely outcomes

One common type of mathematical model of a chance experiment is based on identifying outcomes that are equally likely. For this kind of model, the probability $P(A)$ of an event A is the number of outcomes in A divided by the total number of outcomes in Ω.

Independent events

Events A and B are called *independent* if $P(A \cap B) = P(A) \cdot P(B)$. Durrett gives examples of independent events and explains where this definition comes from using "conditional probability." We will return to conditional probability in a few weeks.

Consequences of the basic rules

Probability of a complement

For any event A, the events A and A^c are disjoint. Therefore, $P(A \cup A^c) = P(A) + P(A^c)$. On the other hand, $A \cup A^c = \Omega$, since every outcome is either in A or in A^c. Therefore, we have $P(A) + P(A^c) = 1$. Solving for $P(A^c)$, we find the following:

$P(A^c) = 1 - P(A)$.

Probability of the event $\varnothing$

We have $\Omega^c = \varnothing$ and $P(\Omega) = 1$. Therefore, we have $P(\varnothing) = 0$.

Probability of a union that is not necessarily disjoint

Let A and B be any events. Durrett explains how to deduce the following formula from the rules of probability:

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

We will soon learn a version of this formula ("inclusion-exclusion") involving more than two events.

If A and B are independent, then so are A and B^c

Let A and B be independent events. We can check that A and B^c are independent as follows:

- $A = (A \cap B) \cup (A \cap B^c)$. You can reason out that equation by convincing yourself that every outcome in A is either in $A \cap B$ or $A \cap B^c$, or you can deduce it from the algebra of events. The two pieces in the union are disjoint, since anything in both pieces is in B and in B^c.
- Therefore, $P(A) = P(A \cap B) + P(A \cap B^c)$.
- Since A and B are independent, we have $P(A \cap B) = P(A) \cdot P(B)$.
- Solving for $P(A \cap B^c)$, we find
  $P(A \cap B^c) = P(A) - P(A \cap B) = P(A) - P(A) \cdot P(B) = P(A) \cdot (1 - P(B)) = P(A) \cdot P(B^c)$,
  which means that A and B^c are independent.

“At least one” for independent events

Let A and B be independent events. The event “at least one of A and B happens” is $A \cup B$. Let’s calculate $P(A \cup B)$ with a version of our “at least one” method from counting:

$P(A \cup B) = 1 – P((A \cup B)^c) = 1 – P(A^c \cap B^c) = 1 – P(A^c) \cdot P(B^c)$.

This formula is helpful, if you know P(A) and P(B), from which you can calculate $P(A^c)$ and $P(B^c)$. To get the formula, we used the following:

- One of De Morgan’s laws from the algebra of events: $(A \cup B)^c = A^c \cap B^c$.
- The fact that A^c and B^c are independent, which follows from the independence of A and B.